

Macro-level 3D memory model — terms

Geometry & counts

C, A, L	capacity, footprint/layer, layers
t_{layer}	physical thickness per layer
P_{max}	power-density budget (cooling)
N_{ind}	indep. peripheral sets
N_{sh}	arrays sharing sense amps
N_{tot}	total arrays = $N_{\text{ind}} N_{\text{sh}}$
$N_{\text{SA}}, N_{\text{dec}}$	# sense amps, # decoders
b_{acc}	bits per access (sense width)
$N_{\text{BL}}, N_{\text{WL}}$	bitlines, wordlines / array
N_{cell}	total cells = $N_{\text{BL}} N_{\text{WL}} N_{\text{tot}}$

Timing & volume

t_{cyc}	steady-state cycle time
$t_{\text{dec}}, t_{\text{WL}}, t_{\text{BL}}$	decode, wordline, BL-develop
$t_{\text{SA}}, t_{\text{sw}}$	sense (incl. restore), selector switch
$\text{Vol}_{\text{arr}}, \text{Vol}_{\text{SA}}$	array, sense-amp volume
$\text{Vol}_{\text{dec}}, \text{Vol}_{\text{sel}}, \text{Vol}_{\text{per}}$	decoder, selector, periph. vol.

Sensing (Upton App. A/B)

Δ, f_{Δ}	settle fraction; factor $-\ln(1-\Delta)$
$a_{\text{lin}}, a_{\text{quad}}$	BL develop lin.; wire self-RC
$r_{\text{BL}}, c_{\text{BL}}$	BL resistance, cap per cell
C_{cell}	storage cap per cell
nU_t, I_{read}	subthreshold; read I (current)
$v_{\text{ratio}}, R_{\text{p-up}}$	divider; pull-up R (voltage)
m_{SA}	SA offset margin (charge-share)

Energy & power

$k_{\text{col}}, k_{\text{arr}}$	energy coeffs $C_{\text{cell}} V^2, c_{\text{BL}} V^2$
V	sense/precharge swing
$e_{\text{periph}}, e_{\text{write}}$	periph., per-cell write energy
f_w, p_{leak}	write fraction; leakage/bit
$E_{\text{acc}}, E_{\text{bit}}$	per-access, per-bit energy
P_{dyn}, A_u	dyn. power; occupied footprint

Cycle time, bandwidth, and packing

Cycle time (throughput bound)

$$t_{\text{cyc}} = \max\left(t_{\text{dec}}, t_{\text{SA}} + t_{\text{sw}}, \frac{t_{\text{dec}} + t_{\text{WL}} + t_{\text{BL}} + t_{\text{SA}} + t_{\text{sw}}}{N_{\text{sh}}}\right)$$

Bandwidth

$$\boxed{\text{BW} = \frac{b_{\text{acc}}}{t_{\text{cyc}}} N_{\text{ind}}} \quad b_{\text{acc}} \leq N_{\text{BL}} \quad (\text{whole-row-activate limit})$$

Counts & volume (packing equality)

$$N_{\text{tot}} = N_{\text{ind}} N_{\text{sh}}, \quad N_{\text{SA}} = N_{\text{ind}} b_{\text{acc}}, \quad N_{\text{dec}} = N_{\text{ind}}$$

$$\text{Vol} = \text{Vol}_{\text{arr}} N_{\text{tot}} + \text{Vol}_{\text{SA}} N_{\text{SA}} + \text{Vol}_{\text{dec}} N_{\text{dec}} + \text{Vol}_{\text{sel}}(N_{\text{sh}}) + \text{Vol}_{\text{per}} N_{\text{tot}} \leq A L t_{\text{layer}}$$

Scaling laws, sensing, and power density

Timing scaling (t_{WL} wire/cell; t_{BL} sense-signal settling, Upton 2024)

$$t_{WL} = k_{WL}^w N_{BL}^2 + k_{WL}^c N_{BL} \quad (\text{WL wire self-RC} + \text{cell/gate load})$$

$$t_{BL} = f_{\Delta} (a_{lin} N_{WL} + a_{quad} N_{WL}^2), \quad f_{\Delta} = -\ln(1 - \Delta), \quad a_{quad} = \frac{1}{2} r_{BL} C_{BL}$$

mode	a_{lin} (signal develop)	note
current (App. A)	$c_{BL} n U_t / I_{read}$	
voltage (App. B)	$v_{ratio} R_{pull-up} C_{BL}$	v_{ratio} also scales a_{quad}
charge-share (DRAM)	$r_{BL} C_{cell}$	row cap $N_{WL} \leq \frac{C_{cell}(1-m_{SA})}{c_{BL} m_{SA}}$

Energy & power density (per-access CV^2 , DESTINY)

$$E_{acc} = k_{col} N_{BL} + k_{arr} N_{BL} N_{WL} + e_{periph} + f_w e_{write} b_{acc}, \quad k_{col} = C_{cell} V^2, \quad k_{arr} = c_{BL} V^2$$

$$E_{bit} = E_{acc} / b_{acc}, \quad P_{dyn} = BW \cdot E_{bit}, \quad A_u = Vol / (L t_{layer})$$

$$(P_{dyn} + p_{leak} N_{cell}) / A_u \leq P_{max}$$